

Q.4 Solve Any Two of the following.**12**

- A) Explain Binary Search Tree. Draw the resulting BST for following data element 20,10,18,4,8,5,13,16,17,1,27

L2/CO4

6

OR

Explain various graph representation methods

- B) Create a binary Tree for given inorder and preorder of a tree

Inorder: E A C K F H D B G

L3/CO4

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Preorder: F A E K C D H G B

- C) a) With reference to Figure-1 define the following,

i) Graph

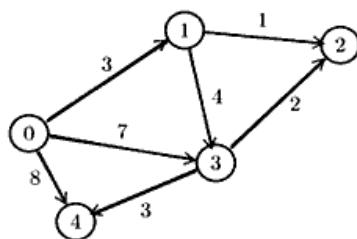
ii) Weighted Graph

iii) Degree of a Graph

iv) Adjacent Vertex

v) Path

vi) Connected Graph



L1/CO4

6**Q. 5 Solve Any Two of the following.****12**

- A) Define algorithm. Explain algorithm analysis with help of worst, average and best case with suitable example.

L2/CO5

6

- B) Write a c program for searching an element in a given array using Binary Search method.

L2/CO5

6

- C) Define sorting. State various sorting technique. Write an algorithm of bubble sort technique.

L2/CO5

6***** End *****